

Total No. of Questions : 8]

SEAT No. :

PC-1687

[Total No. of Pages :2

[6353]-3
T.E. (CIVIL)
DESIGN OF STEEL STRUCTURES
(2019 Pattern) (Semester - I) (301003)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.No. 1 or Q.No. 2, Q.No. 3 or Q.No. 4, Q.No. 5 or Q.No. 6, Q.No. 7 or Q.No. 8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) IS: 800-2007 & IS: 875 Part I to III are permitted.

Q1) a) A column having effective length of 3.5m is subjected to factored axial load of 450 kN and factored moment 50 kNm. Design the column section. Check for section strength only. **[12]**

b) Differentiate between moment resisting base and gusseted base. **[5]**

OR

Q2) a) Design a gusseted base for built up column ISHB 350 @ 67.4 kg/m with plates 450 × 22 mm carrying on axial factored load of 300kN. The column is to be supported on concrete pedestal of M20 grade. Draw the design sketches. **[12]**

b) Explain the following cases with example. **[5]**

- i) Axial force and uniaxial loading
- ii) Axial load and biaxial bending

Q3) a) A simply supported steel joist of 4.0m effective span is laterally supported. It carries a total uniformly distributed load of 40 kN/m inclusive self-weight. Design an appropriate section using I-section. **[10]**

b) Determine the design bending strength of ISMB 400 @ 61.6 kg/m considering the beam to be laterally supported. The effective span of the beam is 3.0 m. Assume steel of grade Fe 410. **[8]**

OR

Q4) a) Calculate safe uniformly distributed load over a laterally unsupported beam ISMB 100 @ 61.6 kg/m for an effective length of 6m. Also check for serviceability. **[13]**

b) Explain factors affecting plastic moment capacity. **[5]**

P.T.O.

Q5) a) Design a gantry supporting an electronically operated crane for following data: [13]

- i) Capacity of crane 120kN
- ii) Span between crane rails 20m Self weight crane girder 100kN
- iii) Weight of crab electric motor hook 15kN
- iv) Minimum hook approach 1.2m
- v) Wheelbase 2m
- vi) Span of Gantry 5.5m
- vii) Weight of rails 0.3kN/m

b) Explain Loading on Gantry girder. [4]

OR

Q6) a) Determine panel point dead load, imposed load and wind load for a truss. Assume design wind pressure as 1200 N/m², use A.C. Sheet and the centre spacing of truss as 6m. Assume 6 panels @5m and Rise 5m. Considered pratt truss. [13]

b) Define: [4]

- i) Pitch
- ii) Central rise.

Q7) A simply supported welded plate girder of an effective span of 20 m subjected to factored uniformly distributed load 40 kN/m throughout the span including the self-weight of plate girder. Assume compression flange laterally supported throughout the span and yield stress of steel is 250 MPa Design cross section of plate girder, stiffeners, and connections. Draw sectional plan and elevation. [18]

OR

Q8) a) Design a welded plate girder for an effective span 28m and carrying a factored UDL of 35kN/m and factored concentrated loads of 140kN acting at 8m from both ends. Design flange and web section. Provided connection between webs and flange. [12]

b) Define plate Girder. Also state it's the element and function. [6]

